

WARP/WARP 12: Navigational Operations Manual

Interstellar Warp Dominoes Federation

Multi-trail Interstellar Dominoes, with federation terminology.

0.0.1 Warp vs Warp 12

Our app was initially only targeted at Double Twelve dominoes and was simply called Warp 12, befitting that scope. We then expanded the domino set capabilities to be able to play double 9 through 18 and in many places the game became known just as Warp or Warp Dominoes.

As an aside, we like Warp 12- it has a certain ring to it.

0.0.2 A note on "official" rules

Unlike chess, multi-trail dominoes originally had no governing body and no single canonical ruleset. Almost every published source, commercial app, and family table differs on the details — hand sizes, the value of the blank, when a train opens, how doubles are handled, and dozens of optional variants. There was no authority to appeal to; there was only common practice.

While Warp does not claim otherwise, we have formed the Interstellar Warp Dominoes Federation to be a fun semi-fictional authority. We have studied the most widely cited sources and tournament-style conventions and tried to stay true to the *spirit* of authentic multi-trail play, while making deliberate, documented choices wherever those sources disagree. The result is meant to be three things at once:

- **A faithful, flexible base.** The most common variations are supported as configurable options, so you can tune a table toward whatever "standard" your group grew up with.
- **An opinionated house standard.** The **Official Warp rules** preset (Section VI) bundles our recommended choices and adds a little fun on top — much as other domino publishers have shaped their own signature editions.
- **A fun, but elevated and refined experience.** Warp combines the frantic action of "Mexican Train" dominoes with the decorum of the space academy and the refined future envisioned by so many science fiction authors.

Where this manual says "standard" or "tournament practice," read it as *the common multi-trail convention Warp has chosen to adopt* for members/players of the IWDF, not a claim of sanctioned authority across "Mexican Train" as a whole. This is our best attempt to honor the game, cover the popular variants, and establish a Warp standard worth playing. This is the IWDF Warp standard, but is by no means an authoritative standard beyond Warp players.

This document is the authoritative rules reference **for Warp** (and Warp only). **Sections I–V** follow widely published multi-trail practice (double-twelve set, engine double set aside before the deal, personal trains, community / Neutral Zone train, train markers, doubles, boneyard, multi-round scoring). **Section VI** lists optional Warp modules and the **Official Warp rules** recommended preset.

Digital implementation note: Warp enforces these rules in software. Setup defaults to **Official Warp rules** (Section VI); hosts may change any toggle before launch. This manual describes behavior when modules are **on** or **off** as stated in each section. **Section VII** describes AI officers and the tactical advisor; **Section VIII** describes solo TEI and the public leaderboard (digital play only). The app also offers **Warp 9 / 15 / 18** as **exhibition** (unrated) sets — see Section II.

0.1 The Captain's Oath — Honor of the Fleet

Warp has no referees. Like the living-room multi-trail tables it descends from, it runs on the honor of the people at the table — and on an older ideal, the one every officer sworn to explore the deep black knows by heart: that *how* you serve matters as much as whether you win.

Every captain who takes the conn is expected to hold the line:

- **We play with Honor.** A clean win is the only win worth logging. We do not cheat, exploit bugs, or manipulate a match to move a rating.
- **We use sanctioned code.** Rated play runs on the official Warp build and its published engine. We do not tamper with the client, spoof results, or automate our turns.
- **We earn our rating.** TEI reflects genuine, unassisted skill. We do not farm it against weak opponents, collude to feed it, or sandbag to protect it.
- **We keep the pool clean.** If we witness cheating, we name it and report it. Guarding the integrity of the leaderboard is every captain's duty, not the fleet's alone.
- **We respect the table.** Opponents, AI officers, and hosts all deserve a fair match, played to its finish. Chatter is kept appropriate for officers on the bridge.

A rated sector is a matter of record — treat it like one. Only sanctioned builds and eligible matches are ever rated; an unrated table carries no standings, but it carries the same courtesy. Play like it matters, because to a captain worth the uniform, it always does.

0.2 Victory conditions

Fleet command chooses one objective before the sector opens:

Mode	Goal
Points campaign (<i>standard</i>)	Thirteen rounds — Spacedock descends from the highest double (eg 12-12) through 0-0 . Lowest cumulative points total wins the campaign.
Go out	First captain to empty their hand in a single round wins the sector immediately (no thirteen-round tally).

0.3 I. Operations lexicon

Multi-trail	Warp
Dominoes	Navigational Coordinates
Engine / station double	Spacedock
Personal train	Warp Trail
Community / public train	Neutral Zone
Train marker	Distress Beacon (<i>Shields Down</i>)
Boneyard	Uncharted Sectors
Open Market (Drafting)	Sensor Grid — optional Module Gamma visible market
—	Red Alert — a double must be satisfied before normal play continues (cover coordinate, or stabilizers when Subspace Fracture applies)
—	Subspace Fracture — optional chicken-foot protocol on doubles (scope: Own Trail, All Captains, or All Doubles)
—	Continuum Flash — optional Module Alpha anomaly on 0-0
—	All Stop! — ceremony required after some round-winning charts (Neutral Zone, or any go-out when All Stop! echo is active)
—	Drop to Impulse — optional house-rule announce when one coordinate remains (uno / knock)
—	Hazard Marker — optional Module Delta hot potato transferred on Neutral Zone contact

0.4 II. Mission setup

Setup matches standard double-twelve multi-trail practice:

1. **Scramble** all 91* coordinates face down.

*Or however many are appropriate for the warp factor in play

1. **Set aside Spacedock** before dealing — **12-12** (round 1), **11-11** (round 2), ... **0-0** (round 13). This coordinate is **not** dealt.
2. **Deal hands** from the remainder:

Captains	Hand size
2–4	15
5–6	12
7–8	10 (<i>default; host may set 11 — see below</i>)

Large fleet hand size (7–8 captains). This is the one setup value where widely published rule sets genuinely disagree: Masters of Games and most modern/commercial sets deal **10**, while Galt (1994) and University Games deal **11**.

Warp defaults to **10** (it leaves a healthier Uncharted Sectors pile — 10 coordinates at 8 captains versus only 2 at 11) and lets the host opt into **11** on the setup screen. It has no effect below 7 captains.

3. **Place Spacedock** in the center.

4. **Uncharted Sectors** — all remaining coordinates face down.

Round starter: Round 1 opener is designated by fleet command (typically the host). Each later round, the starter rotates clockwise. The starter charts first; Spacedock was still set aside before the deal.

Optional modules (Section VI) and **Subspace Fracture** must be agreed before launch. Digital setup defaults to the **Official Warp rules** preset (see Section VI). Hosts can change any toggle before launch.

0.4.1 Exhibition sets (*digital* — *Warp 9 / 15 / 18*)

Sections I–V above describe the **double-twelve** game. The digital app also supports larger and smaller multi-trail sets as **exhibition** sectors — same protocols, different bone count and fleet caps. These never update TEI (Section VIII).

Factor	Set	Coordinates	Fleet	Points campaign
Warp 9	Double-9	55	2–4	10 rounds (9-9 → 0-0)
Warp 12	Double-12	91	2–8	13 rounds (12-12 → 0-0) — <i>rated</i>
Warp 15	Double-15	136	2–12	16 rounds (15-15 → 0-0)
Warp 18	Double-18	190	2–18	19 rounds (18-18 → 0-0)

Hand sizes for Warp 9 / 15 / 18 follow the engine’s set profile. The **7–8 captain 10-vs-11** host choice still applies on Warp 15 / 18 when the fleet is exactly 7 or 8; fleets of **9+** use fixed profile sizes (9 / 8 / 7 / 6 as the table grows). Official-rules presets on exhibition factors use that factor’s full campaign length, not thirteen rounds.

0.5 III. Standard gameplay

Play proceeds clockwise. Each turn, unless a special protocol applies (Red Alert, Subspace Fracture, Continuum Flash resolution), a captain does **one** of the following:

1. **Chart** one legal coordinate, if any exist.
2. **Draw** one coordinate from Uncharted Sectors, then follow the draw rules below.
3. **Deploy a Distress Beacon** — after drawing your one coordinate and still being unable to chart, your shields go down, distress beacon goes up, and your turn ends. (With **Manual shield control**, Section VI, you may also lower shields voluntarily during your turn.)
4. **Pass** only when their Distress Beacon is already active, they still cannot chart, and Uncharted Sectors are empty.

0.5.1 Must play when able

If you hold a coordinate that can be legally charted, you **must** chart one coordinate this turn. You may **not** skip charting while legal moves exist.

0.5.2 Eligible routes

When charting, you may play on:

1. **Your own Warp Trail** (or start it from Spacedock on your first matching play).
2. **The Neutral Zone** — communal line; any captain may start or extend it with a matching end.
3. **Another captain's Warp Trail** — **only while their Distress Beacon is active** (Shields Down).

You may choose **any** eligible route when more than one is legal — you are not required to chart on your own trail.

0.5.3 Opening the round

On the first turn of a round, a captain who can play must open **their own Warp Trail** or the **Neutral Zone** with a coordinate matching Spacedock (e.g. any coordinate containing twelve when Spacedock is 12-12). If they cannot play, they draw once; if the drawn coordinate is playable, they must chart it immediately (choosing any legal route). If they still cannot play, they deploy their Distress Beacon.

0.5.4 Drawing from Uncharted Sectors

When you cannot chart and uncharted coordinates remain, you **must draw** one coordinate.

- If the drawn coordinate can be played, you **must** chart it immediately (any legal route).
- If it cannot be played, you deploy your **Distress Beacon** (shields down) and your turn ends. You draw **one** coordinate per turn — you do not keep drawing, and the distress beacon goes up and shields go down after that single failed draw regardless of how many coordinates remain in Uncharted Sectors.

0.5.5 Distress Beacon (train marker)

Deploy your beacon when — and **only** when — you cannot make a legal chart after drawing, or when the pile is already empty and you still cannot chart.

- **Shields Down:** While your beacon is active, other captains may chart on your Warp Trail.
- **Shields Up:** Your beacon is removed **automatically** when **you** chart on **your own** Warp Trail. Charting on the Neutral Zone or another captain's trail leaves your beacon active.
- **You cannot** deploy a beacon voluntarily while you still have a legal chart elsewhere. (*Standard multi-trail* — no “strategic pass.”) With **Manual shield control** (Section VI), you may drop shields voluntarily after starting your own trail — see that house rule.
- **Passing:** If your beacon is already up, you cannot chart, and Uncharted Sectors are empty, you may **pass** your turn without charting; your beacon stays up.

0.5.6 Red Alert pass

If you cannot satisfy an active Red Alert after drawing (or with an empty pile), you deploy your Distress Beacon and **Red Alert responsibility passes** to the next captain. The alert remains until the double is satisfied.

When **Subspace Fracture** is active (Section VI), “satisfy” means **stabilizers** — not a separate cover coordinate. Pass Red Alert only when you cannot add the next stabilizer.

*(Opt-in exception — Section VI **Pass Red Alert without draw or beacon**: the captain who **charted** the double may pass it on immediately, without drawing or deploying a beacon, but only during **Yellow alert**. Once the alert has passed once, standard rules resume for everyone, including that captain when it comes back around.)*

0.6 IV. Doubles — Red Alert protocol

When a **double** (matching pips on both ends) is charted on any eligible route — your Warp Trail, the Neutral Zone, or an open opponent trail — announce **”Red Alert!”**

- The captain who charted the double must **satisfy** it by playing another valid coordinate on that double **in the same turn sequence**, unless turn rules below apply. *(Standard cover: one matching coordinate on the double. Subspace Fracture: three stabilizers — Section VI.)*
- **Going out:** An empty hand does **not** win the round while your chart left a double unsatisfied. Chart the cover (or final stabilizer) first; only then may an empty hand end the sector.
- If you cannot satisfy it, draw from Uncharted Sectors; if you still cannot, deploy your Distress Beacon and pass Red Alert to the next captain.

Worked example — pass Red Alert after already drawing. Captain A draws once from Uncharted Sectors, then charts a double they cannot cover. Uncharted still has coordinates, but Captain A **does not draw again — one draw per turn**. They deploy their Distress Beacon and **pass Red Alert** to the next captain.

Worked example — empty hand without going out (tournament-style). Four captains, points campaign, Red Alert only (no Subspace Fracture). Captain A's **last coordinate is 12-12**. A charts it on Captain B's open Warp Trail but **does not cover** the double. A's hand is empty, but the round **does not end** — A was **not officially out**.

Red Alert responsibility is on A. A cannot cover the double, so A **passes Red Alert** (drawing first if Uncharted Sectors still hold coordinates and A has not drawn this turn; otherwise deploy beacon and pass). Responsibility moves to B, then C. Captain C **covers** the double with **11-12**. Red Alert clears and normal play resumes.

Captain B still holds **one coordinate** and later charts it legally, emptying B's hand. **B wins the round** and scores **0** for that round. A's hand is still empty, but A is **not** the round winner.

If B had not gone out: play would continue around the table. A is still **not** out of the sector. On A's later turns with **no legal charts** and the beacon up: if Uncharted Sectors are **empty**, A may **pass**; if coordinates **remain**, A **must draw one** before passing (same stuck-with-no-play rule as everyone else) and re-enters the round with a coordinate in hand. The round ends when someone **legally** goes out or the sector is **blocked** (Section V).

Points campaign: A may still win the **sector** on lowest cumulative penalty even when another captain wins this round — **round winner** and **sector winner** are different tallies (Section V).

- While Red Alert is active, **no other routes** may be played until the double is satisfied.

- **Dead double:** If every coordinate in the set containing that pip is already charted (all thirteen coordinates showing that number in a double-twelve set), the double cannot be satisfied — Red Alert ends and normal play resumes.
- Covering a double on another captain's open trail **does not** clear their Distress Beacon.

One double at a time. Warp resolves doubles one at a time: chart a double, satisfy it (cover coordinate, or three stabilizers under Subspace Fracture), and your turn ends. Because a cover must match the double's pip and no other double shares that pip, a cover is always a non-double — you never chain into a fresh Red Alert, and you do not play multiple doubles in a single turn. (Some casual house rules let a captain lay several doubles in one turn as long as the last is covered by a non-double; Warp follows the tournament-style single-double resolution instead.)

0.6.1 Yellow alert and Red Alert (*digital status*)

The rules engine treats every uncovered double as **Red Alert** — the same protocol throughout. On the bridge display and in the sector log, you will see two **status labels** for the same alert:

- **Yellow alert** — the double was just charted; the responsible captain has **not** yet passed Red Alert to someone else.
- **Red Alert** — responsibility has passed at least once (typically after a Distress Beacon deploy and pass). The double still must be satisfied.

This is presentation only. Gameplay, sounds, and scoring follow the Red Alert rules above in both phases.

0.6.2 Subspace Fracture interaction (*Section VI — opt-in*)

When Subspace Fracture is **enabled**, scope is chosen at launch:

Scope	Doubles that open Subspace Fracture
Own Trail (<i>default</i>)	Only on your own Warp Trail
All Captains	On any Warp Trail (yours or an opponent's open trail)
All Doubles	On any Warp Trail or the Neutral Zone

When a double in scope is charted:

1. **Subspace Fracture and Red Alert open together** on that double.
2. While the fracture is open (fewer than three stabilizers), the responsible captain may play **only stabilizers** — not a cover coordinate, not other routes.
3. The **third stabilizer satisfies the double** and clears **both** Subspace Fracture and Red Alert. No fourth coordinate is required on the trail to “cover” the double.
4. If the responsible captain cannot stabilize, draw and/or pass Red Alert as usual; the next responsible captain continues adding stabilizers until all three are placed.

Doubles **outside** the chosen scope never open Subspace Fracture — Red Alert cover rules in this section apply unchanged.

When Subspace Fracture is **off**, all doubles use **Red Alert only** (single cover coordinate).

0.7 V. End of round and scoring

0.7.1 Winning the round

The round ends when one captain charts their **last** coordinate (empty hand) **and** no Red Alert or Subspace Fracture on that chart still requires satisfaction. You **cannot** go out on an open double — cover it (or complete three stabilizers when Subspace Fracture applies) before the sector closes. See the worked example in **Section IV** (empty hand without going out).

All Stop! When the house rule is **on** (default), a round-winning chart on the **Neutral Zone** — or any go-out while **All Stop! echo** is active — ends the sector immediately and the app **automatically** logs and announces **All Stop!** (sound + round log). No helm hold and no manual button.

Turn **All Stop! ceremony** off in game options for pure standard multi-trail play: Neutral Zone go-outs end the round silently, with no announcement.

All Stop! echo (Continuum Flash Module Alpha): when active, **any** round-winning chart triggers the same auto ceremony when the house rule is on.

Drop to Impulse (*house rule — off by default*): when a captain charts down to **one** coordinate remaining, they are **at impulse** — they must call this before their turn completes, and they **cannot chart again** on that turn. This matches standard multi-trail **knock**. (No, reaching one coordinate does not let you silently play that last coordinate.)

They must **Drop to Impulse!** (announce and **pass helm** with one coordinate still in hand) or **pass** without announcing and risk a **catch**. On a **later** turn they may chart their last coordinate under normal rules — including going out when the hand empties.

If they **pass** without having announced, any other captain may **catch** the miss while the catch window is open. A successful catch forces the forgetful captain to draw from Uncharted Sectors (1 or 2 coordinates per game setup, if any remain) — they **return to warp** (no longer at impulse). The window opens when they pass without announcing and **closes** when the **next** captain passes helm (if no one caught). This is separate from **All Stop!** ceremony.

If they **cannot play** their last coordinate and **draw** from Uncharted Sectors while at impulse, that draw **returns them to warp** — there is no need to announce first; drawing already ends the impulse state.

0.7.2 Blocked sector

If Uncharted Sectors are empty and **no captain** can make a legal chart (after dead doubles are resolved), the round ends **without** a domino winner. **Every** captain scores the pip total of coordinates in hand — no exempt captain. All table coordinates, hands, and any remaining pile are shuffled for the next round's deal (standard multi-trail recycle — trails are not trimmed to open ends).

0.7.3 Round scoring (*points campaign*)

When a round ends with a winner, every **other** captain totals pip values in hand and adds that to their campaign score. The round winner scores **0** for that round.

Double-blank (0-0): a double-blank caught in hand scores by the **Double-blank score** setting — **50** (tournament standard, the default for hand-built rule sets), **25**, or **0** (pips). The **Official Warp rules** preset uses **0** so the 0-0 stays safe to hold as the Continuum trigger (Module Alpha). This is independent of the Salamander Penalty, which only affects the **highest double for the Warp Factor** (eg **12-12** under Warp 12, **18-18** under Warp 18).

Next round: Spacedock steps down one double ($12-12 \rightarrow 11-11 \rightarrow \dots \rightarrow 0-0$). After round 13, lowest cumulative points total wins.

0.7.4 How to tally the score (*step by step*)

The app does this automatically at round end and shows a summary before the next deal. The procedure below is for understanding the math (and for tabletop play with physical coordinates).

1. **Decide who counts.** When a captain goes out (empty hand), that **round winner scores 0**. Every **other** captain counts the coordinates left in their hand. In a **blocked sector** (no winner), there is no exemption — **every** captain counts their hand.
2. **Count a hand.** A coordinate is worth the **total number of pips on it — both ends added together**. A blank end counts as **0**. So:
 - $5-3 = 8$, $6-0$ (blank) = **6**, $0-0$ (double-blank) = **0 pips**.
 - A **double counts both halves**: $9-9 = 18$, $12-12 = 24$.
 - Add every coordinate in the hand to get that captain's **round score**.
3. **Apply special-coordinate settings** (if enabled — see Section VI):
 - **Double-blank (0-0):** scored by the **Double-blank score** setting — **50** (tournament standard, the default for hand-built rule sets), **25**, or **0** (its 0 pips). The **Official Warp rules** preset uses **0**.
 - **Salamander Penalty (Module Beta):** a held **highest double** ($\text{maxPip}-\text{maxPip}$) scores **double its both-ends value** from round 2 onward — eg Warp 12 **12-12** $\rightarrow$ **48**, Warp 18 **18-18** $\rightarrow$ **72**. See **Section VI** $\rightarrow$ **Module Beta**.
4. **Add to the campaign total.** Add each captain's round score to their running cumulative total. Lower is better. If **Module Kappa (Temporal Inversion)** is on and this is an **even** round, do **not** use this normal write-up — use the inverted tally in Section VI (spread off the round's top hand; going out takes the full baseline). The campaign still ends on lowest cumulative total.
5. **Advance the round.** Step Spacedock down one double ($12-12 \rightarrow 11-11 \rightarrow \dots \rightarrow 0-0$), re-deal, and play the next round. There are **13 rounds** in a full points campaign.
6. **Declare the winner.** After round 13, the captain with the **lowest cumulative total** wins the sector. If two or more captains tie on the lowest total, they **share** the victory.

Worked example (round 5, Salamander off). Armstrong goes out $\rightarrow$ **0**. Lovell holds $5-3$, $9-9$, $6-0 \rightarrow 8 + 18 + 6 =$ **23**. Earhart holds $0-0$ and $4-2$ with Double-blank score = $50 \rightarrow 50 + 6 =$ **56**. Those round scores are added to each captain's campaign total; Armstrong adds nothing.

Go out mode. There is **no pip tally** — the first captain to empty their hand in any single round wins the sector immediately.

0.8 VI. Optional directives and variants

Warp has designed several Modules (currently 11, Alpha through Lambda) that add optional gameplay mechanics. (**Subspace Fracture** — the chicken-foot protocol on doubles — is part of **core** multi-trail play, not a Module; it is documented as a separate directive below.) We have taken extensive measures to keep rated play unaffected: a skill/luck quantifier and over 171,000

game combinations across module configurations, checking for strange outcomes or behaviors. Some modules are marked **Warped** — intentional chaos that throws the game on its head, leans into luck, or can give an edge to lesser competitors. Those are exhibition-only and never rated.

*Agree before launch. Unless noted, these are **not** part of standard multi-trail tournament practice — though **Sections I–V** (trails, marker, draw, doubles, scoring) still apply underneath.*

0.8.1 Official Warp rules (*recommended preset*)

This is the **encouraged default** for Warp — living-room tables, online sectors, and any future **Warp-branded** play. It is **not** a claim of sanctioned third-party tournament rules; it is the rules bundle this project recommends when you want the full Warp experience without assembling toggles by hand.

In the app, choose **Official Warp rules** on the setup screen (or accept the defaults). Everything in **Sections I–V** applies; the preset only turns on the extras below.

Setting	Official Warp preset
Module Alpha — Continuum	On
Module Beta — Salamander Penalty	On
Drop to Impulse	On — 1-coordinate catch when opponents catch a missed announce
All Stop! ceremony	On — auto log/sound after Neutral Zone wins and All Stop! echo go-outs
Subspace Fracture	Off (hosts may enable separately)
Deluxe house rules	Off (own trail first, NZ after all trails, beacon on any play, round starter plays two)
Manual shield control	Off
Pass Red Alert without draw or beacon	Off
Double-blank (0-0) score	0 (pips) — hand-built/standard rule sets default to 50 (tournament standard)
Objective	Points campaign (thirteen rounds) — <i>Go out</i> remains an optional lobby mode

Hosts may mix and match any toggle; the preset is a one-click reset to the recommended bundle.

0.8.2 Subspace Fracture (*chicken foot — off in Official Warp preset*)

When **enabled**, choose a **fracture scope** before launch:

Scope	Doubles that open Subspace Fracture
Own Trail (<i>default</i>)	Only on your own Warp Trail
All Captains	On any Warp Trail
All Doubles	On any Warp Trail or the Neutral Zone

When a double in scope is charted, **Subspace Fracture** and **Red Alert** open together:

- Fleet navigation halts until the fracture is **stabilized** (three branches from the double).
- While the fracture is open, the captain holding Red Alert responsibility may play **only stabilizers** matching the fracture pip — no cover coordinate, no other routes.
- Each stabilizer must match the double's pip (e.g. any coordinate containing nine on a 9-9 fracture).
- If you cannot add a stabilizer, draw; if still unable, deploy your Distress Beacon and pass Red Alert. The next responsible captain continues stabilizing.
- Layout: the first two stabilizers branch from the double; the **third stabilizer is the center foot** and continues the warp trail from the double.
- The **third stabilizer satisfies the double** — it clears both Subspace Fracture and Red Alert. **No separate cover coordinate** is required afterward.
- When the third stabilizer is placed, normal play resumes across all routes from the center foot's open end.

When Subspace Fracture is **off**, doubles use **Red Alert only** (one cover coordinate on the double).

Fracture immunity (Continuum Flash Module Alpha): the next double on your own trail opens Red Alert but **does not** open Subspace Fracture (regardless of scope).

0.8.3 House rules (*Deluxe-style — off in Official Warp preset*)

Hosts may enable any combination before launch.

Toggle	When enabled
Require own trail first	You must chart at least one coordinate on your own Warp Trail before playing on an opponent's open trail. The Neutral Zone and your own trail are unaffected.
Neutral Zone after all trails started	The Neutral Zone cannot be started until every captain has at least one coordinate on their own Warp Trail.
Beacon clears on any play	Any legal chart removes your Distress Beacon — not only a chart on your own trail.
Round starter plays two	The round starter must chart two coordinate on their own Warp Trail on their opening turn (Spacedock + two from hand). If the first coordinate is a double, the Red Alert cover counts as the second. If you cannot play the second coordinate, deploy your Distress Beacon — no extra draw. Cannot start the Neutral Zone or opponent trails until both are played.
Drop to Impulse	At one coordinate left (at impulse), your turn continues but you cannot chart until you Drop to Impulse! (announce and pass helm) or pass without announcing (opponents may catch — 1 or 2 coordinate penalty per setup). Draw while stuck at impulse → return to warp . Catch window closes when the next captain passes helm.

Toggle	When enabled
Pass Red Alert without draw or beacon	Only the captain who charted the double gets this break, and only while the alert is still in Yellow alert (before it has passed to anyone). If they cannot cover their own double, they may pass responsibility to the next captain without drawing from Uncharted Sectors and without deploying their Distress Beacon. Red Alert then proceeds normally: every other captain — and the original captain when it cycles back to them — must follow standard rules (draw when coordinates remain, deploy the beacon on pass).
Manual shield control	Replaces the automatic "shields rise when you chart your own trail" behavior with manual control, all during your turn (never between turns). Open (Shields down): you may open your own Warp Trail at any time, for any reason — even with legal plays in hand and even before you've started your trail. Close (Shields up): after you drop shields you must chart at least one coordinate on your own Warp Trail before you may raise them again (that own-trail chart may be this turn or a later one); charting your own trail does not auto-raise. One shield change per turn — a single open or a single close, never both and never repeated. Shield changes never pass helm on their own; adjust, then chart or Pass . Standard draw/marker rules still apply when you cannot chart (a forced marker after a failed draw ends your turn).

0.8.4 Module Alpha — The Continuum (*on in Official Warp preset*)

When **enabled**, charting **0-0 on your own Warp Trail** triggers a **Continuum Flash** before helm passes. That captain immediately chooses **one** directive for the rest of the round (cleared when the sector scores):

Continuum Flash	Effect
Reverse turn order	Helm passes counter-clockwise for the rest of the round.
Skip lowest points	The captain with the lowest campaign points score skips their next turn.
Peek Uncharted Sector	The invoker sees the next coordinate that would be drawn from Uncharted Sectors (hidden from others).
Temporal inversion	Turn order reverses until the next double is charted on the table.
Distress amplification	All warp trails are open to every captain without a Distress Beacon.
Fracture immunity	The next double on an own trail will not open Subspace Fracture (Red Alert still applies).

Continuum Flash	Effect
Salamander swap (<i>requires Module Beta</i>)	If anyone holds the highest double (maxPip-maxPip) at round end, the full Salamander penalty (double its pips) lands on the highest-points captain among those who did not win the round instead — the holder pays nothing for that coordinate. If the holder is already that captain, the swap does not transfer: they keep the full doubled penalty (same as Salamander without swap).
All Stop! echo	Any captain going out this round must call All Stop! before the sector closes.
Continuum Wager	Draw two coordinates from Uncharted Sectors — keep one, return the other face-down.

0-0 charted on the Neutral Zone or an opponent's trail does **not** trigger Continuum Flash. A winning 0-0 on your own trail still requires Continuum Flash resolution before the sector can close.

0.8.5 Module Beta — The Salamander Penalty (*on in Official Warp preset*)

If a round ends and a captain holds the **highest double for the Warp Factor** (**maxPip-maxPip**) in hand, that coordinate scores **double its both-ends pip value**. Round 1 never applies (that double is Spacedock). From round 2 onward it is in circulation.

Examples: Warp 12 **12-12** → **48** (base 24); Warp 9 **9-9** → **36**; Warp 15 **15-15** → **60**; Warp 18 **18-18** → **72**.

With **Salamander swap** (Continuum, Module Alpha), the **entire doubled penalty** transfers to the **highest-points captain among non-winners** of the round — the highest-double holder pays **nothing** for that coordinate, and that leader receives the full amount. The round winner is not eligible as the swap target (they score 0 for the round hand). If the holder is already that campaign leader, there is no transfer: they keep the full doubled penalty, exactly as under Salamander without Continuum swap.

0.8.6 Module Gamma — Long-Range Sensor Sweep

High-density sectors demand high-fidelity intel. When DecisionQ metrics spike, captains cannot rely on intuition alone; they require real-time tactical awareness to optimize their navigational throughput.

When **enabled**, navigating the Uncharted Sectors is augmented by a **Sensor Grid** to manage the increased decision density of large-fleet engagements.

- **The Sensor Grid:** A market of 4 to 5 face-up coordinates is dealt face-up next to the Uncharted Sectors.
- **Sensor Sweep vs. Blind Jump:** When a captain is required to draw a coordinate, they may perform a **Sensor Sweep** (select from the Sensor Grid) to maintain optimal decision quality, or execute a **Blind Jump** (draw a random face-down coordinate from the Uncharted Sectors) if the tactical situation requires an uncharted coordinate.
- **Replenish:** Whenever a coordinate leaves the Sensor Grid (Sensor Sweep, or any module that draws through the market), replace it from Uncharted Sectors so the market returns to its configured size while Uncharted still has coordinates. Blind Jumps draw only from Uncharted and do not change the face-up market.

- **Real-Time Updates:** The grid provides a constant, refreshing view of the immediate navigational environment, allowing for precise hand-management in complex, high-skill sectors.

0.9 Module Delta: Hot Potato (Neutral Zone Hazard)

The Neutral Zone is highly unstable subspace. Playing there is convenient but dangerous—every contact transfers custody of a volatile **Hazard Marker**, and captains stuck holding it when they pass suffer immediate penalties.

0.9.1 The Hazard Marker (Hot Potato)

Transfer Rule: Whenever a captain charts a coordinate onto the **Neutral Zone**, they immediately take custody of the **Hazard Marker**.

Pass Penalty: If you **pass** your turn while holding the Hazard Marker (cannot chart, drew or pile is empty, beacon already deployed), you suffer a **+5 point penalty** added to your round score.

- The penalty applies **each time** you pass while holding the marker
- Multiple passes in a round stack: 2 passes = +10 points, 3 passes = +15 points, etc.
- The marker stays with you until someone else plays to the Neutral Zone

Round Initialization: The round starter receives the Hazard Marker at the beginning of each round (they got the “privilege” of the opening coordinate).

Warp Drive Engagement (Spool): On your turn (when not blocked by Red Alert or Subspace Fracture), you may **engage warp drive** to draw coordinates from Uncharted Sectors—draw coordinates one at a time until you draw a coordinate that cannot be charted to your chosen route. All matching coordinates are charted; the mismatch (and any remaining coordinates) go to your hand. Spooling is part of Module Delta; Module Theta does not unlock it. Engage Warp Drive is unavailable when Uncharted Sectors (and the Sensor Grid, if Module Gamma is on) have no coordinates left to draw.

Tactical Impact: The Neutral Zone becomes a strategic dumping ground with real risk. Captains must weigh convenience against the danger of getting stuck with the marker and being forced to pass. Spooling stretches a trail (or the Neutral Zone) at hand-bloat risk.

0.9.2 Scoring

When the round ends, each captain's score is:

$$\text{Final Score} = \text{Hand Pips} + \text{Pass Penalties}$$

Pass Penalties: +5 points for each time you passed while holding the Hazard Marker during this round.

Worked Example (Module Delta enabled, 3 captains, Points campaign):

Round 5 ends. Captain Armstrong goes out (empty hand). Captain Lovell and Captain Earhart remain:

- **Lovell's hand:** [5-3], [9-9], [6-0] = 8 + 18 + 6 = **32 pips**
- **Lovell passed once with marker:** +5 penalty
- **Lovell's round score:** 32 + 5 = **37**

- **Earhart's hand:** $[0-0], [4-2], [11-8] = 0 + 6 + 19 = \mathbf{25 \text{ pips}}$ (assuming Double-blank score = 0)
- **Earhart never held marker / never passed:** +0 penalty
- **Earhart's round score:** $25 + 0 = \mathbf{25}$
- **Armstrong (went out):** 0 round score

Armstrong adds 0, Earhart adds 25, Lovell adds 37 to their respective campaign totals.

0.9.3 Module Theta: Longest Trail Bonus

Strategic long-range navigation rewards extended personal trails. Captains who invest in building the longest continuous warp trail receive a tactical advantage at round end.

Longest Trail Bonus: At round end, the captain(s) with the longest personal Warp Trail receive a **-3 bonus** (negative = fewer points). If multiple captains tie for longest, all tied captains receive the bonus. The bonus **cannot reduce a captain's round score below 0** (a go-out or tiny hand stays at 0 after all normal-round adjustments). Engaging Warp Drive / Spool is **not** part of Theta—it ships with Module Delta.

Tactical Impact: Creates opposing incentives with Module Delta (hot potato and NZ spool push toward Neutral Zone risk; longest trail pulls toward your own trail).

Status: Rated — preserves skill ordering (78.5% higher-skill win rate in calibration).

0.9.4 Module Iota: Double Down

Doubles become tactical weapons. When you chart a double, the next captain must immediately draw additional coordinates—forcing hand bloat on opponents.

Mechanic: When any captain charts a double to any eligible route (own trail, Neutral Zone, or opponent's open trail), the **next captain** draws **2 coordinates** from Uncharted Sectors (or Sensor Grid if uncharted exhausted) before their turn begins.

Tactical Impact: Doubles shift from liability (Red Alert) to weapon (burden next player). Encourages double hoarding or strategic timing. Adds variance through forced draws.

Status: Rated — excellent skill preservation (81.4% higher-skill win rate in calibration).

0.9.5 Module Eta: Temporal Debt

Drawing from the unknown carries consequences. Each time you reach into Uncharted Sectors, you accumulate Temporal Debt—a penalty due at round's end.

Mechanic:

- Each blind draw from **Uncharted Sectors** accumulates **+1 debt token**
- At round end, each debt token adds **+2 points** to your final score (regardless of whether you went out or were caught holding)
- Debt tokens **reset to zero** at the start of each new round
- **Sensor Sweep draws (Module Gamma)** do NOT accumulate debt—you're drawing from visible market, not blind chance

Example: You make 3 blind draws from uncharted during the round. At scoring, you're caught with 18 pips in hand + 3 debt tokens ($\times 2$) = **24 points total**.

Tactical Impact:

- **Punishes excessive drawing** — the classic “draw until you can play” strategy becomes costly
- **Rewards playing from hand** — skilled coordinate management (holding versatile coordinates, planning ahead) gains an edge
- **Strategic drawing** — you must weigh: is drawing worth +2 penalty? Can I play from hand instead?
- **Sensor Grid synergy** — Module Gamma's visible market becomes even more valuable (no debt penalty for targeted draws)

Status: Rated — preserves skill ordering by rewarding hand management and strategic restraint (calibration pending).

0.9.6 Module Kappa: Temporal Inversion (*Warped — exhibition only*)

Reality bends. Who “wins” a round flips every other round, but the **campaign still ranks by lowest cumulative total**. Kappa does not replace that ledger — it only changes how each even round is written into it.

Which rounds invert:

- **Odd rounds** (1, 3, 5...): use the normal tally in Section IV (empty hand = 0; others add hand pips; lower is better).
- **Even rounds** (2, 4, 6...): use the inverted tally below.

Even-round tally (tabletop): Count hands the same way as Section IV (pip totals, double-blank setting, Salamander if on). Find the **top hand** — the largest hand pip total any captain still holds this round. Then write each captain’s **campaign adjustment** as a *spread* off that top hand — do **not** award “0 for going out” on these rounds.

1. **Baseline = top hand.** Every captain starts the round at that number.
2. **Subtract your own hand.** The captain holding the top hand nets **0** — they win the round. A thinner hand nets the difference (top hand minus theirs). Whoever **emptied their hand** (went out) kept nothing, so they take the **full baseline** — the worst outcome, since on an inverted round you wanted to *keep* coordinates, not shed them.
3. Apply any other enabled Module add-ons from this section (Hazard pass penalties, Longest Trail −3, Temporal Debt, and so on) **after** that hand term, same as the app.
4. The sector winner at the end is still whoever has the **lowest cumulative** score — Kappa never switches the campaign to “highest wins.”

Scoring floor and ties (both round types): The lowest score a captain can *show* is **zero**. The only term that can dip a round below zero is the Longest Trail bonus (−3); that sub-zero amount is **kept** so it can break an otherwise-tied campaign (a true −3 edges a 0), but it is never *displayed* as a negative — scoreboards clamp the shown total to zero and use the raw value only to rank level totals.

Why the top hand wins: On an even round a fat hand should *help* you under a lowest-total campaign, and the spread does exactly that: the biggest hand contributes 0, thinner hands contribute the gap, and an empty hand (a go-out) contributes the most. Unlike the old fixed dump, the swing scales with the actual hands — one bad even round no longer dwarfs an entire campaign. Offline, just keep a running total column and add each captain’s spread as written.

Worked example (Warp 12, even round, three captains). Before the round: Armstrong 20, Lovell 25, Earhart 30. Earhart goes out; Armstrong holds 40 pips; Lovell holds 12. The **top hand is 40** (Armstrong).

- Armstrong (40 in hand): $40 - 40 = +0 \rightarrow$ campaign $20 + 0 = \mathbf{20}$ — wins the round with the fattest hand.
- Lovell (12 in hand): $40 - 12 = +28 \rightarrow$ campaign $25 + 28 = \mathbf{53}$.
- Earhart (went out): $40 - 0 = +40 \rightarrow$ campaign $30 + 40 = \mathbf{70}$.

Armstrong leads the campaign for holding the most; Earhart is penalized for emptying out when coordinates were the prize — but by a proportional 40, not a flat 312.

Tactical Impact: Going out in even rounds is the worst outcome (you take the full top-hand baseline). Medium-to-large hands become optimal. AI struggles to adapt. Lower-skilled players win ~65% of the time (35.6% higher-skill win rate in calibration — intentionally inverts skill ordering).

Status: Exhibition/Warped mode only—**intentionally breaks skill ordering** for casual chaos play. Never rated (TEI does not use these scores).

0.9.7 Module Lambda: Wormholes (*Warped — exhibition only*)

Spatial instability permits topological manipulation. Captains can open **Wormholes** that transpose their personal Warp Trail with the Neutral Zone, fundamentally altering board state ownership.

Mechanic:

- **Wormhole Trigger:** Playing a **double** onto the **Neutral Zone** opens a Wormhole
- **Spatial Inversion:** The captain's personal trail and the Neutral Zone **immediately swap ownership**
 - Your former trail becomes the new public Neutral Zone (open to all)
 - The former Neutral Zone becomes your new private trail
- **Distress Beacon:** If you had a beacon active before the swap, it is **destroyed during transit** (fresh start)
- **Red Alert Resolution:** The double triggers Red Alert on your **newly acquired** trail. You must satisfy it from your hand; if you cannot, you deploy a beacon on your new trail.

Strategic Implications:

- **The Escape Pod:** Dump a dead-end trail full of blocked doubles into the public sphere and claim the Neutral Zone's momentum
- **Risk Assessment:** Before opening a Wormhole, ensure you can answer the Red Alert after the swap; otherwise you beacon your stolen trail open to everyone

Tactical Impact: Fundamentally rewrites the game from linear coordinates shedding to aggressive board state hijacking. High-skill players' long-term trail building becomes a vulnerability (theft target). AI heuristics must weigh theft opportunities against defense. Lower-skilled players benefit from the randomness of dynamic ownership (~50-60% expected skill preservation).

Status: Exhibition/Warped mode only—**predicted to significantly reduce skill ordering** (untested; awaiting calibration). Never rated.

0.9.8 Module Epsilon — Tactical Requisition (The Captain's Draft)

In top-tier Warp 18 engagements, victory is determined at the drafting table. Pre-mission optimization is essential; captains must requisition their tactical loadout to match the projected skill ceiling of the sector.

When **enabled**, the random blind deal is replaced by a drafting phase, allowing captains to engage in strategic loadout optimization before the sector launch.

1. **Requisition Packs:** Uncharted Sectors are scrambled. Each captain receives a Requisition Pack.
2. **First Selection:** Captains analyze their initial intel, selecting one coordinate to lock into their permanent tactical loadout.
3. **Data Transfer:** Remaining coordinates are passed clockwise, forcing captains to adapt their strategy based on the available data pool.
4. **Strategic Depth:** This phase transforms the start of the sector from a luck-based event into a high-skill predictive challenge, ensuring that every captain enters the sector with a coherent, optimized navigation strategy.

0.9.9 Module Zeta — Fleet Squadrons (Warp Crews)

The DecisionQ requirement of 18-captain sectors is extreme. To maintain peak operational performance, the fleet forms localized task forces, pooling their cognitive resources to manage the massive navigational complexity of the board.

When **enabled**, the fleet divides into equal squadrons before the sector launches. Squadrons are the professional standard for managing high-complexity 18-captain sectors.

- **Squadron Formation:** Captains divide into equal teams. This structure allows for shared cognitive load and collaborative high-level strategy.
- **Bridge Seating:** Teammates alternate turns with opposing squadrons to ensure balanced tactical pressure.
- **The Shared Trail:** Each squadron shares a single Warp Trail and a single Distress Beacon, creating a unified navigational front.
- **Shield Protocols:** The squadron's shields remain "Up" as long as *any* member of the team maintains forward momentum on their shared trail.
- **Collaborative Command:** Teammates may openly discuss high-level strategy and coordinate moves, effectively leveraging shared brainpower to master the complex decision trees of the Warp 18 board.
- **Scoring:** Victory is a team achievement; a squadron goes out when *one* captain clears their hand, with the team penalty calculated by the aggregate remaining pips.

0.10 VII. AI officers & tactical advisor (*digital*)

Local simulation and online sectors can fill empty chairs with **AI officers**. The in-game **tactical advisor** uses the same decision stack to suggest moves and explain reasoning. This section describes what those systems know and how **Lookahead** works.

0.10.1 What an AI officer can see

An AI captain has the same **public** information you do:

- Its **own** hand (coordinates held).
- Every coordinate already **charted** on the table (all Warp Trails, Neutral Zone, fracture stabilizers).
- How many coordinates each opponent is **holding** (hand count only — not which coordinates).
- How many coordinates remain in **Uncharted Sectors**.

An AI officer **cannot** see which specific coordinates are in an opponent's hand. It does not read hidden coordinates from the game state when choosing a move.

0.10.2 AI officers (Ensign / Lieutenant / Commander)

Everyone at the table is a **Captain** (seat role). **Ensign**, **Lieutenant**, and **Commander** are opponent / placement tracks — commission flavor for AI difficulty, not a separate ladder from TEI:

Track	Profile
Ensign	Provisional / new profile — more blunders, lighter heuristics
Lieutenant	Competent / standard
Commander	Veteran / sharp — self-play neural policy (Ω): greedy inference, ms/move. Same Commander / σ = commander label — not a separate “Commander+” tier.

Controls how often the officer blunders and how sharply it prefers high-scoring lines. All tracks still obey the same legal-move and rules-engine constraints.

0.10.3 Lookahead (per-officer opt-in)

When **Lookahead** is **off**, the officer uses a fast **greedy** policy: score each legal move for this turn with heuristics (shed heavy pips, own-trail pressure, Red Alert safety, objective mode, modules, and so on) and pick among them.

When **Lookahead** is **on**, the officer **forward-searches** before acting:

1. For each candidate move, it runs the move through the **real Warp rules engine** a few turns ahead.
2. Because opponent hands and the draw order are hidden, each simulation **guesses** plausible holdings: opponents receive a random assignment from the pool of coordinates not on the table and not in the AI's hand, while preserving each opponent's **actual hand count**. Uncharted Sectors are filled from the same unseen pool.
3. The search repeats that guess several times (**determinizations**), averages the outcomes, and picks the move that tends to work best across those possible worlds.

Lookahead is **imperfect-information search** — not clairvoyance. It is slower but can reason about consequences (for example, whether a play sets up an opponent to go out). Commission track still applies on top (blunders and noisy tie-breaking).

0.10.4 Tactical advisor

The tactical advisor suggests one move plus plain-language reasons so humans can see *why* a line is strong, not only what to play.

Today: it uses the **Commander** simulation profile with **Lookahead** enabled (heuristic stack).

Planned: a hybrid neural–heuristic model **distilled from** Ω — trained to agree with the Commander neural officer's picks and explain them in named game concepts. It does **not** imitate legacy Commander heuristics. Assisted play remains unrated (below).

Invoking the tactical advisor **during live play** marks the match as **assisted** for rating purposes (Section VIII). Post-match advisor reports and campaign downloads do **not** affect whether a match is rated.

0.10.5 Ω + extended thinking (*planned, exhibition*)

The same Ω neural weights can run with **extended thinking** — Commander-free PUCT search (Omega policy prior + value leaves) with a per-move iteration budget. Stronger than greedy Commander play when the value head is competent; slower. Intended for exhibition / casual hard mode, not as a separate rated anchor.

0.11 VIII. TEI & leaderboard (*digital*)

Local solo sectors against **AI officers** can report results to **iwdf.org** when the client is signed in to Firebase. Team campaigns, unrated lobby modes, and builds without a working stats backend do not update the public boards.

In-person play: While we encourage Warp to be played in person with physical dominoes — around a table, with friends and family — the TEI rating system requires digital tracking to compute and maintain ratings accurately. For tabletop games, use the TEI calculator or officiated match codes to report results and update your rating. The full online app (warp.iwdf.org) and the leaderboards (iwdf.org) handle all TEI computation automatically.

Warp keeps **three rating contexts**, each split into **Go-out** and **Points** tracks:

- **Solo pool** — unassisted matches against reference AI officers (Ensign / Lieutenant / Commander).
- **Human pool** — online sectors and officiated matches **without** a crew charter (global `humanTei`).
- **Crew charters** — friend-group ladders on iwdf.org/crews; scoped **group TEI** per charter. See Crews & charters.

TEI is displayed as a **grade** (letter + score, like “V67”) combining your skill estimate and rating confidence. For how that grade relates to the underlying OpenSkill model, see the in-app page **How TEI works** (linked from Profile and About), or the normative write-up in `docs/tei-spec.md`.

0.11.1 Online sectors (human pool)

A completed online sector is rated into the **human pool** when it meets every condition below; otherwise it plays out normally but changes no ratings. The lobby shows a live **rated / unrated** banner so the fleet knows before launch.

- **The set is double-twelve (Warp 12).** Warp 9 / 15 / 18 are **exhibition** — the lobby forces unrated and the server rejects TEI reports for those sets.
- **Two or more captains are signed in with an account.** Guests (anonymous sign-in) can play, but a guest at the table makes the whole sector unrated — a rating that can't persist across devices isn't a rating.
- **Every AI officer is Ensign / Lieutenant / Commander.** These have fixed reference strength and serve as rating **anchors**: finishing above or below them moves your TEI, which keeps online results on the same scale as solo play. An experimental **Class I*** search officer makes the sector unrated. (Commander is a neural policy — still rated, same σ =commander anchor.)
- **The objective is Points or Go-out.** Other modes are unrated.
- **No captain consulted the tactical advisor.** Just as in solo play, invoking the in-game advisor during live turns makes a match *assisted* — and online, one assisted captain leaves the **whole** sector unrated. You are warned the moment you engage it.

When the sector ends, every signed-in captain's TEI is updated with a single pairwise pass over the final standings (see the spec, §6.5 / §10). The server re-derives the result from the authoritative game record and re-checks each seat, so no captain can report a score they didn't earn. Ratings are applied once per sector.

0.11.2 Crews & charters (group TEI)

A **crew** is a persistent friend group with a **charter** — a frozen contract: Official Warp rules (warp12-official-v1), objective (Points or Go-out), fleet size (2–8), and campaign length. When a rated match or online sector is played **under a crew charter**, TEI updates go to that crew's ladder (groupTei), not the global human pool.

- **Create / join** at iwdf.org/crews. **Google sign-in required.** No shared crew passwords — invite links only (owner can rotate).
- **Officiated nights:** the match official selects the crew when creating an MT- code. All checked-in captains must be crew members. Approval moves **crew TEI only**.
- **Online sectors:** the host selects a crew in the lobby waiting room; fleet size, objective, and rules lock to the charter. Sector must match exactly or it stays unrated (**charter_mismatch**).
- **Guests** at the table do not earn crew TEI (same as global human pool).
- **Global Official** is a special open charter (**global-official**). Rated play under it updates **both** the Global Official crew ladder **and** global human-pool TEI.

The TEI calculator can preview crew-ladder outcomes without saving anything. Full design: [docs/crews-roadmap.md](#).

0.11.3 Lexicon

Term	Meaning
Captain	Seat at the table — every player is a Captain
TEI (Tactical Efficiency Index)	Your rating, displayed as a grade (letter + score, like “V67”)
TEI Grade	Letter indicating rating confidence: E (Elite), V (Veteran), C (Consistent), I (Improving), P (Provisional)
TEI Score	0–99 number showing conservative skill on one global scale — not restarted per letter
Commission track	Opponent / placement tier on file: Ensign , Lieutenant , Commander . Personal federation commission (Cadet–Fleet Admiral) is derived from your TEI. Flag Officer prestige covers Rear Admiral and above

0.11.4 Federation commission (ranks)

Your TEI maps to a **federation commission** — Cadet through Fleet Admiral — so a grade like “V67” also reads as a naval rank on the HUD and profile. It is **flavor over TEI**, not a second rating: same grade string, same update rules.

Seat vs commission: every player at the table is still a **Captain** (seat role). Commission ranks never replace that word.

Opponent tracks vs personal commission: rated AI buckets and Academy placement use three coarse tracks — **Ensign** / **Lieutenant** / **Commander**. Your *personal* commission is finer (Cadet through Fleet Admiral) and updates whenever your TEI does.

Lieutenant Junior Grade (Lt. JG) is the naval rung between **Ensign** and full **Lieutenant** — an early Improver who has cleared Ensign territory but is not yet mid-trail Lieutenant strength. The app may show **Lt. JG** in tight HUDs; tooltips and this manual use the full name.

Commission	Short	Typical TEI band
Cadet	Cdt.	Below P25
Ensign	Ens.	P25–I25
Lieutenant Junior Grade	Lt. JG	I25–I40
Lieutenant	Lt.	I40–C45
Lieutenant Commander	Lt. Cmdr.	C45–C55
Commander	Cmdr.	C55–V63
Commodore	Cdore.	V63–V70
Rear Admiral	R. Adm.	V70–V80
Vice Admiral	V. Adm.	V80–V90
Admiral	Adm.	V90–V99
Fleet Admiral	F. Adm.	V99 (and top Elite scores)

Bands follow letter order **P** < **I** < **C** < **V** < **E**, then score. **Elite (E)** uses the same score thresholds as **Veteran (V)** for commission (so E73 sits with high admiralty). **Rear Admiral** and above are **Flag Officer** prestige territory — earned through TEI, never selected at Academy.

0.11.5 Two independent tracks

Your solo TEI is **not** one number — the app keeps separate ratings for each **objective** and each **AI commission track** you face:

Track	When it applies
Go-out TEI	First captain to empty their hand wins the sector
Points TEI	Lowest cumulative points total when the campaign ends (<i>or the round, in single-round solo</i>)

Each track is further split by opponent profile: **Ensign**, **Lieutenant**, and **Commander**. Beating Commander officers does not move your Ensign bucket.

0.11.6 Federation Academy placement

Before your first rated match in each track, the app asks for a **commission track** separately for **go-out** and **points**. Choose **Ensign**, **Lieutenant**, or **Commander** on each track (with a short self-description), then fine-tune a **starting TEI** within that track's band. You might place as Commander for go-out and Ensign for points. Each track is saved **once**; after you save placement for a track — or after your first unassisted rated match in that track — its starting TEI field locks. **Flag Officer** is not selectable at onboarding — it is earned.

Track	Self-description	Points TEI band	Go-out TEI band	Example
Ensign	New to dominoes	P0–I25	P0–I25	P0, I9
Lieutenant	Knows multi-trail	I25–C45	I25–C45	C21, C30
Commander	Seasoned strategist	C45–V70	C45–V70	V53, V62

0.11.7 Fixed opponent reference ratings

Unassisted matches are scored against **fixed** reference ratings — not the other chairs' live TEI. AI officers have calibrated skill estimates that serve as stable anchors:

Track	Ensign	Lieutenant	Commander
Points	Conservative baseline	Competent player	Sharp strategist
Go-out	Conservative baseline	Competent player	Sharp strategist

Your TEI grade reflects how you perform relative to these anchors. Consistently beating Commander officers will push your rating into Veteran (V) or Elite (E) territory, while struggling against Ensign keeps you in Provisional (P) or Improving (I) grades. The leaderboard also shows **percentile** (top X%) within each board so rank stays meaningful across the rating distribution.

0.11.8 How your TEI moves

After each **rated** match, your TEI grade updates to reflect both **how well you play** and **how confident the system is** about your skill level.

Your TEI is shown as **two parts**:

- **Letter grade (E/V/C/I/P)** — How established / confident the rating is. More consistent rated play over time tends to tighten confidence.
- **Number score (0–99)** — Conservative skill on one global scale. Same number $\approx$ same skill even across different letters.

The Letter Grade (Confidence):

Your letter shows how **established** your rating is — how sure the system is that the score is in the right neighborhood. More rated play usually tightens confidence, but a new module or long break can widen it again:

- **P (Provisional)** — Still establishing; high uncertainty.
- **I (Improving)** — Estimate is moving; not locked in yet.
- **C (Consistent)** — Reliable enough for everyday comparison.
- **V (Veteran)** — Highly reliable estimate.
- **E (Elite)** — Tightly anchored rating.

The Number Score (Skill):

Your score is **conservative skill on one global 0–99 scale**. Same number $\approx$ same skill even across letters — **I40** and **C40** are equally strong estimates; C is simply more established. The score stays deliberately cautious: until confidence rises, the displayed number sits a bit below a raw “best guess.”

- **Win games** against strong opponents $\rightarrow$ score tends to go up
- **Lose games** against weaker opponents $\rightarrow$ score tends to go down
- **Play more rated matches** $\rightarrow$ confidence tends to rise, which often lifts the conservative score as well

Placement, not margin. Ratings track *where* you finish, not *by how much*. When a match is scored, the final standings are collapsed to finishing places (1st, 2nd, 3rd...) and your rating moves on those places alone: every captain you finish ahead of nudges you up, every one who finishes ahead of you nudges you down, each weighted by how strong that opponent was. The size of the point gap never enters the math — losing a points campaign by 3 and losing it by 300 move your TEI by **exactly the same amount**, and a blowout win counts no more than a one-point squeaker. (Tie on the cumulative total and you split the place, a half-step each.) So a single ugly round — say a Module Kappa inversion or a fat go-out penalty that dumps a big number on you — only costs rating if it actually drops you *past* another captain; if you still finish in the same place, your TEI is unchanged. Chase **finishing order**, not a pretty scoreline.

Two ways to improve your TEI:

1. **Play more rated matches** so confidence tightens (letter tends to climb $P \rightarrow I \rightarrow C \rightarrow V \rightarrow E$)
2. **Win more games** so skill climbs (number toward 99)

Example progression:

- **Week 1:** P15 · Cadet (still learning)
- **Week 3:** I28 · Lieutenant Junior Grade (settling in)
- **Month 2:** C42 · Lieutenant (stable estimate)
- **Month 4:** V58 · Commander (highly reliable)
- **Year 1:** E76 · Rear Admiral (anchored elite confidence)

What happens when you try new modules?

When you play with a new module or rule variant, your **letter grade may temporarily drop** (e.g., V→I) and the conservative score may dip with it. This is intentional — the system is **re-evaluating** your skill under new conditions, not punishing you. Play more games with that module and confidence climbs back once the estimate settles.

0.11.9 Starting TEI (*Academy*)

Before your **first rated** match in each track, complete **Federation Academy placement** for that track: pick Ensign, Lieutenant, or Commander, then save a starting TEI within that track's band. Tracks are independent — strong at go-out but new to points campaigns is fine. If you skip placement and play unassisted, the first match in that track's bucket begins from the **default starting rating** (P25 — Ensign commission floor).

0.11.10 What counts as rated

Solo pool — only **unassisted double-twelve** matches update TEI:

- **Rated:** Warp 12, played without invoking the in-game **tactical advisor** during live turns.
- **Assisted:** you requested a coach suggestion during play. The win or loss still appears in your profile and general stats, but **TEI does not move**. Assisted wins are tracked separately (`advisorMatches` / `advisorWins`).
- **Exhibition:** Warp 9 / 15 / 18 never update TEI, with or without the advisor.

Downloading a post-match advisor report or campaign analysis does **not** disqualify a match.

Human pool — an online sector is rated when it clears the eligibility bar above (Warp 12, two or more signed-in captains, only Ensign / Lieutenant / Commander AI, Points or Go-out, **and no captain used the advisor**). Any guest, experimental Class I* seat, exhibition set, or advisor consult leaves the whole sector unrated.

Crew charter — same eligibility bar, plus: host attached a `charterId`; sector settings match the charter (fleet size, objective, campaign length, rules profile); every human captain is a signed-in **member** of that crew. Private crew matches update crew TEI only. **Global Official** also updates global human-pool TEI.

Commander is neural Ω . Greedy `createOmegaPlayer` is the Commander officer (local, online host, pass-and-play). TEI still uses σ =`commander`; publish `warp12-official-v2` when recalibrated REF-TEI ships. Ω + extended thinking (PUCT) is unrated exhibition. See `docs/omega-handoff.md` and `docs/tei-spec.md` §7.1.3.

0.11.11 After the sector

When a rated match completes, the sector summary shows your TEI grade **before** → **after**.

What the display shows:

- “**I52 → I55 (+3)**” — Your score went up by 3 points (same letter grade)
- “**I55 → C55**” — Your letter improved from I to C! (more experience = more reliable)
- “**V72 → V68 (-4)**” — Your score dropped 4 points after a loss (letter unchanged)

Reading a TEI grade:

- **V67** = Veteran (experienced, reliable rating) with skill score 67 out of 99
- **C42** = Consistent (established rating) with skill score 42 out of 99
- **I35** = Improving (still learning, fewer games) with skill score 35 out of 99
- **P28** = Provisional (new player, rating not yet stable) with skill score 28 out of 99

Want more details? Hover over your TEI grade in the app to see:

- How many rated games you've played
- Your exact skill level and confidence values
- What your grade means

If Firebase is unavailable, local decision-quality feedback may still run, but your TEI won't be saved to the leaderboard.

0.12 IX. Subspace messaging (*digital* — *online sectors*)

Online sectors include a persistent comms channel — **subspace messaging** — so captains can coordinate before launch, react during play, and debrief afterward. Messaging is designed to preserve the integrity of rated sectors while keeping casual games social.

0.12.1 Comms modes

Context	Mode	What's available
Lobby (rated or casual)	Full	Quick-phrase hails, free-form text, DMs
Active play — casual sector	Full	Quick-phrase hails, free-form text, DMs
Active play — rated sector	Quick-only	Public quick-phrase hails only. No free text, no DMs.
Post-game (rated or casual)	Full	All comms re-open once standings are final

The restriction during rated active play exists to prevent collusion — free text or private messages between opponents could be used to coordinate play, which would undermine TEI. The quick-phrase catalog is intentionally social and expressive, never strategic.

0.12.2 Quick-phrase hails

Five groups, each with a category icon:

- **Acknowledge** — Aye Captain · Acknowledged · Make it so · Course laid in
- **Get moving** — Engage! · Punch it · Warp speed · Ahead full · Steady as she goes
- **Sportsmanship** — Well played · A fine maneuver · The needs of the many... · Fly well, Captain
- **Drama** — Red Alert! · Shields up! · All Stop! · Distress beacon away · She's breaking up! · Fascinating... · Bold. Very bold.
- **Cheeky** — You're playing a dangerous game · Persistence is futile · Resistance is... noted · I have the conn now · Q would be proud

Phrases are broadcast publicly and logged in the sector record. They cannot convey hand information or coordinate play, only camaraderie and table talk.

0.12.3 Free-form text and DMs

Available in the lobby, casual active play, and after the sector completes. Messages are visible to all sector members (including DMs — transparency over privacy, consistent with The Captain's Oath). A per-recipient picker lets you direct a message, but every captain at the table can read it.

0.12.4 Moderation

- **Per-user mute:** hide a captain's messages for the remainder of the session.
- **Rate limiting:** a cooldown prevents message flooding (burst of 3, then 3 seconds between sends).
- Future: report action tied to The Captain's Oath's "we identify cheaters and bad actors."

0.12.5 Rated sector toggle

The host can opt out of rating at any time before launch by unchecking **Rated sector** in the lobby. When unchecked:

- TEI will not change regardless of the sector's outcome.
- Full comms (text + DMs) remain available during active play.
- The lobby and bridge show a "Casual sector" banner so all captains know the match is unrated.

0.13 Quick reference — standard vs Warp extras

Official Warp rules (recommended preset) — see **Section VI** — enables Continuum, Salamander, Drop to Impulse (1-coordinate catch), and All Stop! ceremony on standard **Sections I–V** gameplay.

Rule	Standard multi-trail	Warp
Train marker when stuck	Required	Distress Beacon — same
Voluntary marker while able to play	No	No (unless Manual shield control — own trail required)
Play elsewhere while marked	Allowed; marker stays	Same
Play on own trail while marked	Marker must come off	Shields Up — same
Doubles	Must satisfy / cover	Red Alert — same
Chicken foot on doubles	Optional house variant	Subspace Fracture — opt-in; scope: Own Trail / All Captains / All Doubles
Own trail before opponents	Optional Deluxe variant	House rule — opt-in
NZ after all trails	Optional Deluxe variant	House rule — opt-in
Beacon clears on any play	Optional Deluxe variant	House rule — opt-in
Round starter plays two	Optional Deluxe variant	House rule — opt-in
0-0 anomaly	—	Continuum — Official Warp preset

Rule	Standard multi-trail	Warp
Highest-double Salamander Penalty	—	Salamander — Official Warp preset
NZ win announcement	—	All Stop! ceremony — Official Warp preset
One coordinate left announce	—	Drop to Impulse — Official Warp preset (1-coordinate catch)
Double-blank (0-0) score	50 (tournament standard)	Setup option 50 / 25 / 0 — Official Warp preset uses 0 (Continuum trigger)
Blocked boneyard	Round ends, all score	Blocked sector — same
AI officers / tactical advisor	—	Section VII — digital only
Solo TEI vs AI	—	Section VIII — iwdf.org; unassisted matches only
Online TEI (human pool)	—	Section VIII — auto-rated when all captains are signed in and any AI are Ensign-Commander
Crew / charter TEI	—	Section VIII — group TEI via leaderboard crews + officiated or online rated sectors
Global Official TEI	—	Section VIII — open charter; updates crew + global human pool
Subspace messaging	—	Section IX — quick hails always; free-form/DMs in lobby, casual, and post-game only